Building Multi-Step Arguments from Givens

ordering the steps, segment arguments, and angle arguments

- An argument starts at a *given* and ends at the claim. Every line in between must follow from a line already written, by a definition, a postulate, or a property of equality.
- Some problems ask you to order and justify the steps; others ask you to run the chain and compute. Both need the reasons written down.
- Diagrams show labelling only; all measurements are in the problem text.

1. The four statements below make a valid argument, but they are out of order. Number them 1 to 4 in a working order and write the reason for each.

___ $m\angle FGH = 180 - m\angle HGK$

___ $\angle FGH$ and $\angle HGK$ form a linear pair

___ $\overrightarrow{GF}$ and $\overrightarrow{GK}$ are opposite rays, and $\overrightarrow{GH}$ is a third ray from G

___ $m\angle FGH + m\angle HGK = 180$

2. The five statements below prove that a midpoint cuts a segment in half, but they are out of order. Number them 1 to 5 and write the reason for each.

___ $AM = MB$

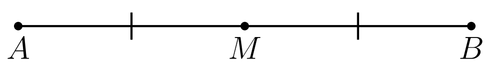
___ $AM + MB = AB$

___ M is the midpoint of $\overline{AB}$

___ $2\,AM = AB$, so $AM = \frac{1}{2}AB$

___ $\overline{AM} \cong \overline{MB}$

3. M is the midpoint of $\overline{AB}$. In centimetres, $AM = 4x - 1$ and $MB = 2x + 7$.



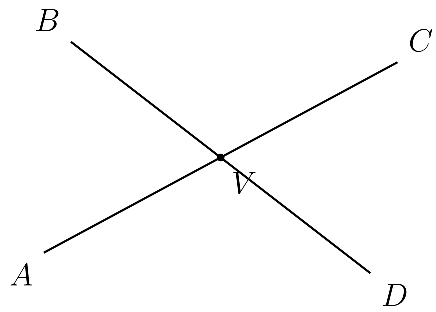
(a) Find x.

Answer: ____________________

(b) Find the whole length AB.

Answer: ____________________ cm

4. $\overleftrightarrow{AC}$ and $\overleftrightarrow{BD}$ intersect at V. The angles $\angle AVB$ and $\angle BVC$ form a linear pair, and in degrees $m\angle AVB = 3x + 15$ and $m\angle BVC = 5x + 5$.



(a) Find x.

Answer: ________________

(b) Find $m\angle CVD$, the angle vertical to $\angle AVB$.

Answer: ________________ degrees

5. The statements below prove that two angles complementary to the same angle are congruent. They are out of order. Number them and write the reason for each.

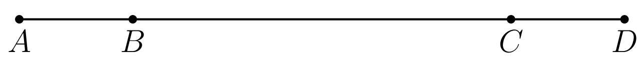
___ $m\angle P = m\angle R$, so $\angle P \cong \angle R$

___ $m\angle P + m\angle Q = m\angle R + m\angle Q$

___ $\angle P$ and $\angle Q$ are complementary; $\angle R$ and $\angle Q$ are complementary

___ $m\angle P + m\angle Q = 90$ and $m\angle R + m\angle Q = 90$

6. Four points A, B, C, D lie in a straight line in that order, and the two outer stretches are equal: $AB = CD$.

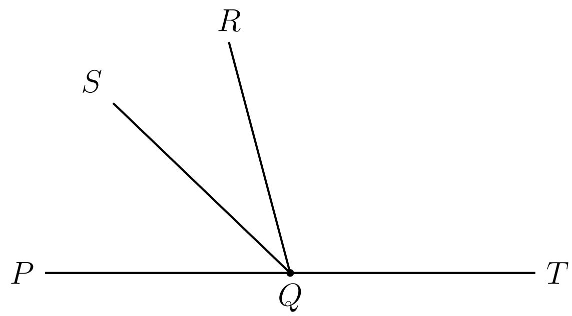


(a) Build an argument that $AC = BD$. Write three steps, each with its reason.

(b) The stretches are then measured: $AB = CD = 4.2$ cm and $BC = 11$ cm. Find AC.

Answer: ________________ cm

7. $\overrightarrow{QS}$ bisects $\angle PQR$, and $\angle PQR$ and $\angle RQT$ form a linear pair with $m\angle RQT = 110$ degrees. In degrees, $m\angle PQS = 2x + 5$.

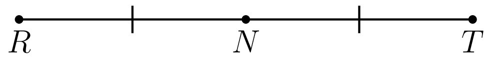


(a) Write the chain of reasons that connects the given 110 degrees to an equation in x.

(b) Find x.

Answer: ________________

8. N is the midpoint of $\overline{RT}$. In centimetres, $RN = 3x + 2$ and $NT = x + 10$.



(a) Find x.

Answer: ________________

(b) Find RT.

Answer: ________________ cm

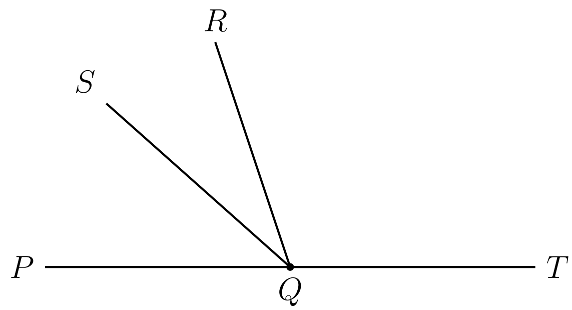
9. $\angle J$ and $\angle K$ are complementary, and $\angle L$ and $\angle K$ are complementary. In degrees, $m\angle J = 4x - 6$ and $m\angle L = 2x + 18$.

(a) Build an argument that $\angle J \cong \angle L$. Write four steps, each with its reason.

(b) Use that result to find x.

Answer: ________________

10. $\overrightarrow{QS}$ bisects $\angle PQR$, and $\angle PQR$ and $\angle RQT$ form a linear pair. In degrees, $m\angle PQS = 2x + 15$ and $m\angle RQT = 6x$.



(a) Build the whole argument, from the two givens to a single equation in x. Write each statement with its reason.

(b) Find x.

Answer: ________________